Visualizing P2P ICP Solving Steps (TikZ Series)

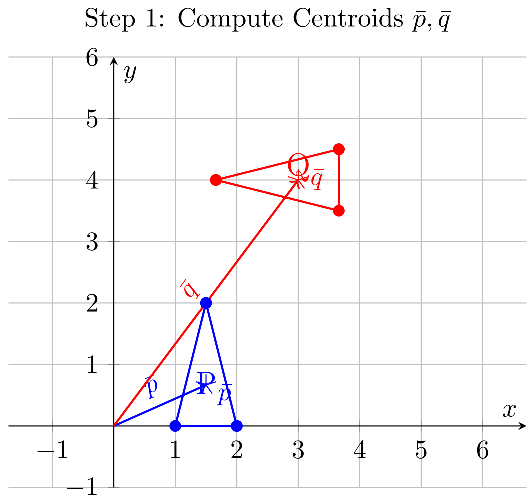

Figure 1: Initial Point Clouds P (blue, offset triangle) and Q (red, offset and rotated triangle), with their calculated centroids $\bar{p}$ and $\bar{q}$ marked and vectors from the origin shown.

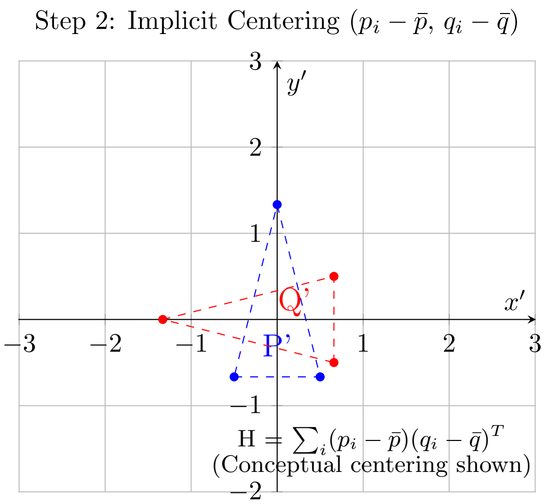

Figure 2: Visualization of Step 2 conceptually centering both point clouds at the global origin to compute the cross-covariance matrix H using the relative positions of points to their respective centroids.

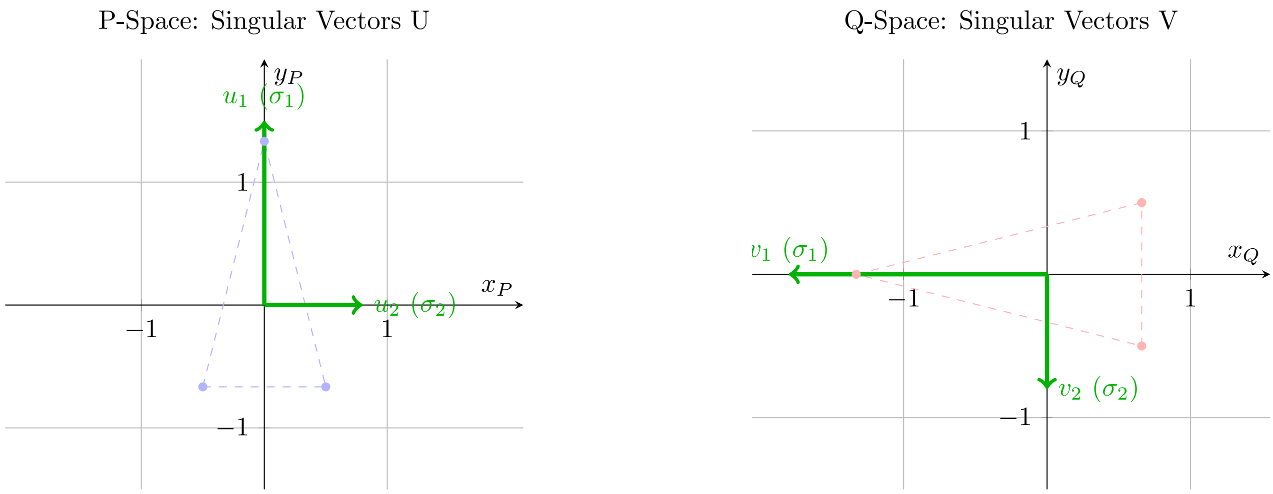

Figure 3: Visualization of Step 3 SVD decomposition $H = U\Sigma V^T$, showing the orthogonal singular vectors U and V as principal axes in the centered frames of reference for P and Q respectively, with illustrative singular values σ_i indicating the scale of variance along these directions.

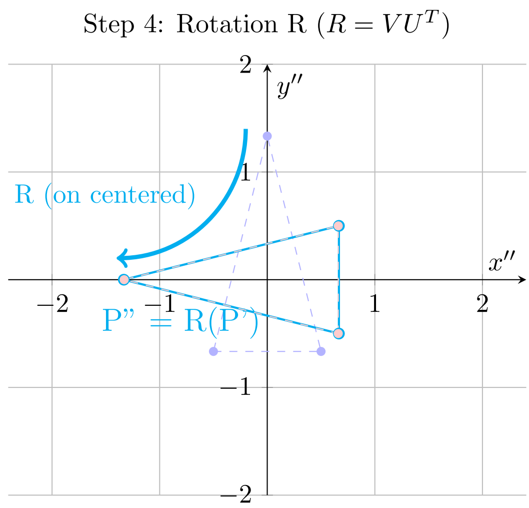

Figure 4: Visualizing Step 4 rotation part, showing how the rotation matrix R is conceptually applied to the centered point cloud P' to align its principal axes with those of centered Q', resulting in the rotated, centered shape P".

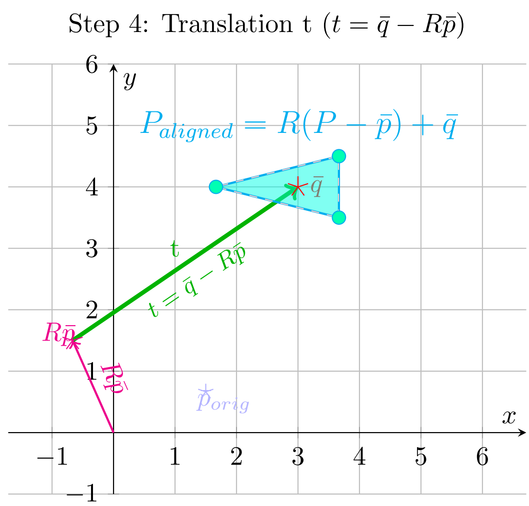

Figure 5: Visualizing the Step 4 translation vector calculation t = $\bar{q} - R\bar{p}$. The vector t shifts the rotated source centroid $R\bar{p}$ directly to the target centroid $\bar{q}$. Conceptual perfect shape alignment (cyan filled triangle) is shown centered on $\bar{q}$, aligning with the original Q shape.